The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.2 (Conservation Ledger and Safe Movement Law)

Status: engine hardened and re-tested. This version adds an explicit conservation ledger and a safe discrete movement law, then reports a three-model comparison, a phase map, grid-size invariance, and a supply-shock recovery test. All claims below are backed by the accompanying test suite and experiment scripts.

Headline results. (1) The safe rule — gradient proposal plus an exact discrete acceptance check and a line-searched transfer size — improves on the raw gradient rule overall (lower burden and unmet demand) and is provably unable to increase burden within a tick; it is not, however, a strict improvement on every metric. (2) On a well-mixed field it holds ~99.4% of cells viable with mean burden ~0.004, approximately size-insensitive across the sizes tested. (3) Homeostasis is not a point but a bounded phase region requiring supply matched to leakage. (4) When sources are clustered away from demand, both rules essentially fail (~51% viable, barely above no-rule) — a failure whose cause (locality, delay, and dissipation) is separated by the V2.2.1 audit and not yet attributed to any single factor.

Abstract

V2.1 showed the potential-gradient movement law could sustain a distributed field indefinitely, but relied on informal accounting and on a marginal (linear) acceptance rule that can overshoot. V2.2 hardens the engine before adding any new theory. Every tick now balances an explicit conservation ledger, dX = S + G - D - Lambda - transport_loss - spill, with unmet demand recorded separately; the identity is asserted at every step and over 500-tick runs. The movement law is made safe: the gradient only proposes a direction, and a transfer executes only if an exact discrete calculation confirms B_with < B_without, with the transfer size chosen by a line search that minimises the pair's combined burden. Seven new hardening tests pass (fifteen tests in total). We then compare three models — no redistribution, raw gradient, and safe — on checkerboard, random, and clustered worlds, across grid sizes, and under a supply shock, and summarise outcomes as a phase map over supply ratio and leakage.

1. Purpose and Relationship to Earlier Versions

V2.0 specified the model; V2.1 implemented it, corrected the leakage term to be state-dependent, and gave a first empirical verdict. V2.2 does not add new physics. Following the project discipline — harden the accounting and the movement law until they are unquestionably correct before introducing regeneration or EBU accounting — it strengthens two things: the flow accounting (a conservation ledger) and the decision rule (a safe discrete acceptance test with adaptive transfer size). Actors remain stationary local processes, closest to circulation in a body.

2. V2.2 Corrections

2.1 Conservation ledger

Each tick records every flow separately and checks the continuity identity (Law 2) exactly. Amounts are the realised values: a cell cannot leak or consume more than it holds, and capacity above K spills:

```
dX  =  S + G - D - Lambda - transport_loss - spill
unmet_demand  recorded separately (demand that could not be met)

per cell:  pre = x + s + g
           actual_leak   = min(leak, pre)
           actual_demand = min(d, pre - actual_leak)
           unmet         = d - actual_demand
           spill         = max(0, (pre - actual_leak - actual_demand) - K)
```

A unit test asserts dX_realised equals the ledger sum every tick and cumulatively over a run (verified: 500 ticks, X change -67.80 == ledger dX -67.80).

2.2 Safe discrete movement law

The potential gradient F_ij = mu_i - mu_j - theta_ij still selects a candidate direction, but it no longer sets the transfer amount and no longer authorises execution. Two safeguards are added.

Adaptive size (line search). The transfer amount minimises the pair's combined burden along the transfer direction (a golden-section search on the convex piecewise-quadratic penalty), instead of the raw q = M[F]_+ which overshoots:

```
q*  =  argmin_{0 <= q <= q_hi}  [ ell_i(x_i - c0 - q) + ell_j(x_j + eta_ij q) ]
q_hi = min( q_a^max,  x_i - x_i^min - c0,  (K_j - x_j)/eta_ij )
```

Exact acceptance. The transfer executes only if it strictly reduces burden on the live state (V2.0 Sec. 9 with H=1):

```
execute  iff   B_with_action  <  B_without_action   -   epsilon
```

Because every accepted transfer strictly reduces B and rejected ones leave the state unchanged, redistribution can never increase B within a tick. Hence Impact = B_without - B_with >= 0 by construction — a provable discrete-monotonicity guarantee, confirmed empirically (worst redistribution impact over 500 dynamic ticks: +0.00e+00). The trade-off versus strict simultaneous resolution is that transfers are applied greedily, strongest force first; each is re-verified against the live state, so the guarantee holds regardless of order.

3. Verification

All fifteen tests pass: the eight V2.0/V2.1 physics tests (bounds, conservation, loss accounting, finite and regenerative sources, the ideal-flow monotonicity proof, reproducibility, counterfactual separation) and the seven V2.2 hardening tests below.

#	V2.2 hardening test	Result
1	Overflow above K recorded as spill; x capped; ledger balances	PASS (spill=5.0, x=20)
2	Demand beyond capacity recorded as unmet; x floors at 0	PASS (consumed 3, unmet 7)
3	Ledger balances every tick and cumulatively (500 ticks)	PASS (dX = -67.80)
4	Harmful gradient proposal is rejected by the safe rule	PASS (grad -21.56 vs safe +1.00)
5	Discrete monotonicity: safe B never increases (100 ticks)	PASS (B -> 0.0000)
6	Redistribution impact >= 0 in the full dynamic world	PASS (worst +0.0)
7	0 <= x_i <= K_i under the safe rule (500 ticks)	PASS

4. Experiments

4.1 Three models on three worlds

A fixed total supply budget (supply_ratio = 2.0 x total demand, distributed among producers) so that only spatial structure differs; kappa = 0.02, eta = 0.95, n = 10, 2000 ticks.

World	Model	viable %	mean B	transport loss	unmet	regime
checkerboard	none	50.0	1040.0	0	39599	-
checkerboard	gradient	94.6	0.330	2582	0	homeostatic
checkerboard	safe	99.4	0.004	2694	0	homeostatic
random	none	47.0	1073.6	0	41975	-
random	gradient	63.7	200.0	10808	1248	deficit
random	safe	81.5	9.81	4604	0	deficit
clustered	none	49.0	1051.2	0	40391	-
clustered	gradient	51.6	550.3	9201	14762	deficit
clustered	safe	51.0	427.5	7952	7332	deficit

The safe rule gives lower burden and unmet demand overall, and higher viability in the mixed worlds. It is not a strict Pareto improvement, however: on the clustered world its viability is essentially tied with the gradient rule (51.0% vs 51.6%), and on the checkerboard it uses slightly more transport (2694 vs 2582). Structure dominates outcome: on the well-mixed checkerboard the safe rule is excellent; on a random field it rescues most cells but ~18% still starve locally; with clustered sources both rules are barely better than doing nothing (~51% vs 49%).

4.2 Grid-size invariance

On the checkerboard the safe rule is approximately size-insensitive across the three sizes tested: viability stays ~99.5% and mean burden stays <0.01 for n = 6, 10, and 14 (36, 100, 196 cells). This is evidence of size-insensitivity over the tested range, not a proof of scale invariance in general.

4.3 Phase map

Sweeping supply ratio against the leakage coefficient shows homeostasis is a bounded region, not a single tuned point. Too little supply gives deficit collapse; too much supply (or too little leakage to absorb it) gives excess accumulation; a diagonal band between them is homeostatic. No oscillation regime appeared under the safe rule in this range — the acceptance safeguard makes the dynamics notably stable.

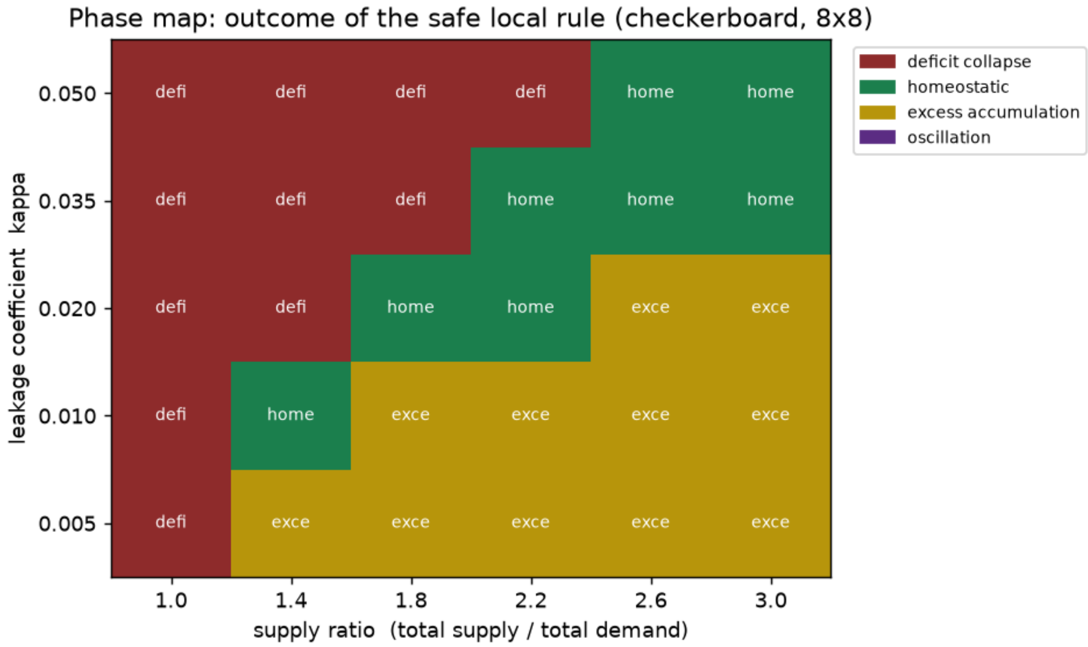

Figure 1. Phase map of the safe rule (checkerboard, 8x8). Homeostasis (green) requires supply matched to leakage; it is flanked by deficit collapse (red) and excess accumulation (gold).

4.4 Recovery after a supply shock

Inflow is set to zero for ticks 400–600. With no redistribution the field never recovers; the gradient rule recovers to 90% viability by tick 671; the safe rule recovers fastest and tightest, by tick 632. All models fall to zero viability during a total supply cut — correctly, since there is nothing to redistribute.

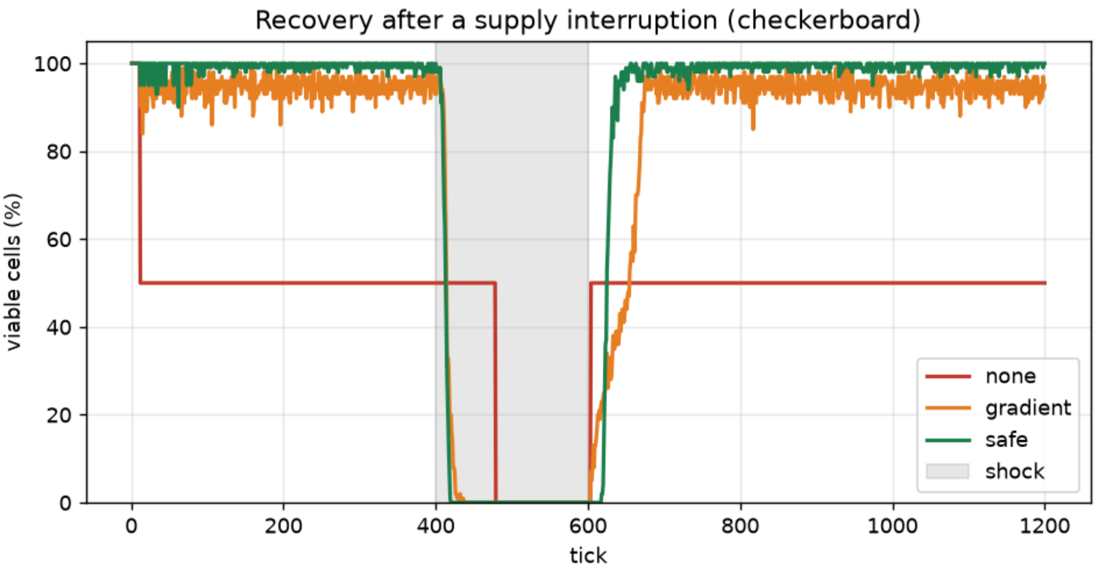

Figure 2. Viable-cell fraction through a supply interruption (grey band). Safe (green) recovers fastest; gradient (orange) is noisier and slower; no-rule (red) does not recover.

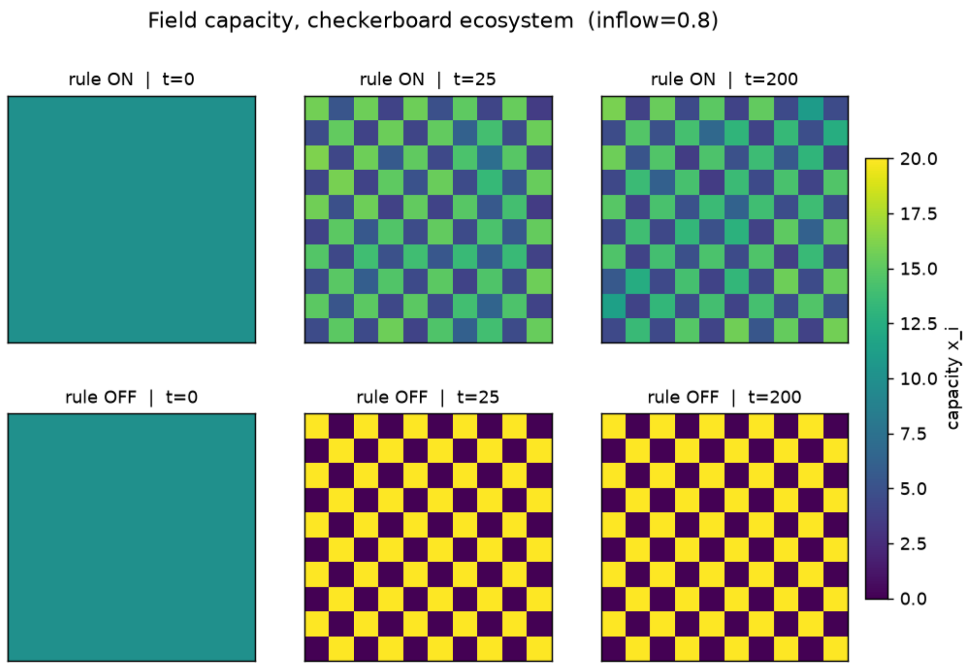

Figure 3. Field capacity on the checkerboard (from V2.1). Top: the local rule holds a balanced field. Bottom: without it, producers pin at K and consumers die at 0.

5. Verdict

The V2.2 engine is trustworthy, and the safe movement law is an overall improvement. The ledger makes every tick auditable and is asserted to balance; fifteen tests pass, including the ideal-flow monotonicity proof and the new discrete-monotonicity guarantee. The safe rule cannot make the field worse within a tick, rejects harmful overshoot that the raw gradient executes, and gives lower burden and unmet demand overall. It is not a strict Pareto improvement: clustered viability is statistically tied with the gradient rule, and it uses slightly more transport on the checkerboard.

But the model does not guarantee survival, and now shows exactly when it fails. Re-checking the falsification criteria (V2.0 Sec. 13):

Criterion (Sec. 13)	V2.2 finding	Status
Persistent shuttling / oscillation with no benefit	No oscillation regime under the safe rule; stable	Not triggered
Transport loss exceeds homeostatic benefit	Apparent for clustered sources (~51% viable); cause not yet separated from myopia/distance	Under audit
Burden hides local collapse behind aggregate	Guarded: per-cell viability + unmet demand now reported	Mitigated
Small coefficient change reverses conclusions	Outcome is a bounded phase region in (supply, kappa)	Characteris ed
Counterfactual costs more than it detects	H=1 acceptance is cheap; line search ~24 evals/transfer	Acceptable
Short horizon destroys long-horizon regeneration	Not yet testable (no regenerative/Allee sources)	Open
EBU incentives cause gaming	EBU not implemented	Open

Bottom line. With the ledger and the safe rule, the movement law is a correct and honest model of local-rule homeostasis. It answers the central question conditionally: purely local, potential-driven, safety-checked transfers do sustain a distributed field indefinitely — provided (a) total supply is matched to total loss, and (b) sources are spatially close to demand. The clustered-source failure is the most valuable result, but its cause is not yet isolated: it may stem from locality, from the delay of moving capacity through neutral intermediate cells under a one-tick (H=1) acceptance rule, from transport dissipation, or from a combination. The V2.2.1 audit (multi-seed layouts, an eta=1 lossless test, and H=3/10/30 foresight) is designed to separate these causes before any conclusion is drawn.

6. Limitations and Next Steps

1. Introduce regeneration properly — logistic resources, a minimum-regeneration (Allee) threshold, and irreversible stocks — then implement the H-tick regeneration-aware actor (Model D) and test whether short-horizon action destroys long-horizon regenerative capacity (the project's central conjecture).
2. Add long-range transport variants and a distance/terrain-dependent theta to map exactly where the clustered-source failure begins.
3. Sensitivity sweeps on eta, M, theta, alpha, beta to complete the phase characterisation.
4. Only after the physics is trusted, add the EBU ledger as an optional overlay and test for gaming and repeated credit.

Version status. V2.2 keeps the V2.0 one-field formulation and seven hard laws, and the V2.1 state-dependent leakage. It adds an audited conservation ledger and a provably safe discrete movement law. The next revision should follow only after regeneration and the H-horizon actor (Model D) are implemented and tested.